Jonah Kudler-Flam

Contact

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Employment

Marvin L. Goldberger Member, Institute for Advanced Study, Princeton, NJ 2022 - 2027
Affiliated PCTS Postdoctoral Fellow, Princeton University, Princeton, NJ 2022 - 2025

Education

University of Chicago 2017 - 2022
M.S. in Physics
Ph.D. in Physics
Advisor: Shinsei Ryu

Princeton Univerity 2021 - 2022
Visiting Student Research Collaborator
Trailing Student

Massachusetts Institute of Technology 2020 - 2021
Visiting Student
Host: Hong Liu

Colgate University 2013 - 2017
A.B. in Astronomy/Physics *magna cum laude*

King's College London 2016
Study Abroad Student

Publications

1. **J. Kudler-Flam**, E. Witten, "Wormholes and Averaging over N" arXiv:2605.15180 [hep-th].
2. M. S. Klinger, **J. Kudler-Flam**, G. Satishchandran, "Generalized Entropy is von Neumann Entropy II: The complete symmetry group and edge modes" arXiv:2601.07910 [hep-th]. Under Review.
3. Akash Vijay, Ayush Raj, **J. Kudler-Flam**, Benot Vermersch, Andreas Elben, Laimei Nie, "Analytically Continuing the Randomized Measurement Toolbox" arXiv:2511.02912 [quant-ph]]. Under Review.
4. **J. Kudler-Flam**, E. Witten, "Emergent Mixed States for Baby Universes and Black Holes" arXiv:2510.06376 [hep-th]. nder Review.
5. A. Herderschee, **J. Kudler-Flam**, "Stringy algebras, stretched horizons, and quantum-connected wormholes " arXiv:2510.01556 [hep-th]. Under Review.
6. A. Blommaert, **J. Kudler-Flam**, E. Y. Urbach, "Absolute entropy and the observer's no-boundary state"arXiv:2505.14771 [hep-th]. Accepted To Journal of High Energy Physics.
7. **J. Kudler-Flam**, V. Narovlansky, N. Sopenko, "Optimal bound on long-range distillable entanglement " arXiv:2504.02926 [quant-ph]. Phys. Rev. Lett. 135, 060403.
8. **J. Kudler-Flam**, G. Penington, "It costs nothing to teleport information into a black hole" arXiv:2504.01058 [hep-th]. International Journal of Modern Physics D. Awarded second prize in 2025 Gravity Research Foundation Essay Contest.

9. **J. Kudler-Flam**, K. Prabhu, G. Satishchandran, “Vacua and infrared radiation in de Sitter quantum field theory ” arXiv:2503.19957 [hep-th]. Under Review.
10. D.L. Danielson, **J. Kudler-Flam**, G. Satishchandran, R.M. Wald, “How to Minimize the Decoherence Caused by Black Holes” arXiv:2501.04773 [hep-th]. Phys. Rev. D 112, 025012
11. H.J. Chen and **J. Kudler-Flam**, “Free Independence and the Noncrossing Partition Lattice in Dual-Unitary Quantum Circuits” arXiv:2409.17226 [hep-th]. Phys. Rev. B 111, 014311
12. X. Dong, **J. Kudler-Flam**, and P. Rath, “Entanglement Negativity and Replica Symmetry Breaking in General Holographic States” arXiv:2409.13009 [hep-th]. Journal of High Energy Physics 01 022 (2025)
13. **J. Kudler-Flam**, S. Leutheusser, and G. Satishchandran, “Algebraic Observational Cosmology” arXiv:2406.01669 [hep-th]. Under Review.
14. **J. Kudler-Flam**, S. Leutheusser, A. A. Rahman, G. Satishchandran, and A. Speranza, “A covariant regulator for entanglement entropy: proofs of the Bekenstein bound and QNEC” arXiv:2312.07646 [hep-th]. Phys. Rev. D 111, 105001
15. X. Dong, **J. Kudler-Flam**, and P. Rath, “A Modified Cosmic Brane Proposal for Holographic Renyi Entropy” arXiv:2312.04625 [hep-th]. Journal of High Energy Physics 06 120 (2024).
16. **J. Kudler-Flam**, S. Leutheusser, and G. Satishchandran, “Generalized Black Hole Entropy is von Neumann Entropy” arXiv:2309.15897 [hep-th]. Phys. Rev. D 111, 025013
17. **J. Kudler-Flam**, M. Nozaki, T. Numasawa, S. Ryu, and M.T. Tan, “Bridging two quantum quench problems – local joining quantum quench and Möbius quench – and their holographic dual descriptions,” arXiv:2309.04665 [hep-th]. Journal of High Energy Physics 08 213 (2024).
18. **J. Kudler-Flam**, L. Nie, and A. Vijay, “Rényi mutual information in quantum field theory, tensor networks, and gravity”arXiv:2308.08600 [hep-th]. Journal of High Energy Physics 06 195 (2024).
19. G. Cipolloni and **J. Kudler-Flam**, “Non-Hermitian Hamiltonians Violate the Eigenstate Thermalization Hypothesis,” arXiv:2303.03448 [cond-mat.stat-mech]. Phys. Rev. B 109, L020201 *Editor’s Suggestion*.
20. Y. Liu, **J. Kudler-Flam**, and K. Kawabata, “Symmetry Classification of Typical Quantum Entanglement,” arXiv:2301.07778 [cond-mat.mes-hall]. Phys. Rev. B 108, 085109.
21. Y. Liu, Y. Kusuki, R. Sohal, **J. Kudler-Flam**, and S. Ryu, “Multipartite entanglement in two-dimensional chiral topological liquids,” arXiv:2301.07130 [cond-mat.str-el]. Phys. Rev. B 109, 085108.
22. **J. Kudler-Flam** and Y. Kusuki, “On Quantum Information Before the Page Time,” arXiv:2212.06839 [hep-th]. Journal of High Energy Physics 08 189 (2022).
23. **J. Kudler-Flam** “Rényi Mutual Information in Quantum Field Theory,” arXiv:2211.01392 [hep-th]. Phys. Rev. Lett. 130, 021603.
24. G. Cipolloni and **J. Kudler-Flam** “Entanglement Entropy of Non-Hermitian Eigenstates and the Ginibre Ensemble,” arXiv:2206.12438 [cond-mat.stat-mech]. Phys. Rev. Lett. 130, 010401.
25. **J. Kudler-Flam** and P. Rath, “Large and Small Corrections to the JLMS Formula from Replica Wormholes,” arXiv:2203.11954 [hep-th]. Journal of High Energy Physics 08 189 (2022).
26. S. Vardhan, **J. Kudler-Flam**, H. Shapourian, and H. Liu, “Mixed-state entanglement and information recovery in thermalized states and evaporating black holes,” arXiv:2112.00020 [hep-th]. Journal of High Energy Physics 01 064 (2023).

27. Y. Liu, R. Sohal, **J. Kudler-Flam**, and S. Ryu, “Multipartitioning topological phases by vertex states and quantum entanglement,” arXiv:2110.11980 [cond-mat.str-el]. Phys. Rev. B 105, 115107.
28. S. Vardhan, **J. Kudler-Flam**, H. Shapourian, and H. Liu, “Bound entanglement in thermalized states and black hole radiation,” arXiv:2110.02959 [hep-th]. Phys. Rev. Lett. 129, 061602.
29. **J. Kudler-Flam**, V. Narovlansky, and S. Ryu, “Negativity Spectra in Random Tensor Networks and Holography,” arXiv:2109.02649 [hep-th]. Journal of High Energy Physics 02 076 (2022).
30. **J. Kudler-Flam**, V. Narovlansky, and S. Ryu, “Distinguishing Random and Black Hole Microstates,” arXiv:2108.00011 [hep-th]. Phys. Rev. X Quantum 2, 040340.
31. **J. Kudler-Flam**, R. Sohal, and L. Nie, “Information Scrambling with Conservation Laws,” arXiv:2107.04043. SciPost Phys. 12, 117 (2022).
32. **J. Kudler-Flam**, “Relative Entropy of Random States and Black Holes,” arXiv:2102.05053 [hep-th]. Phys. Rev. Lett. 126, 171603.
33. H. Shapourian, S. Liu, **J. Kudler-Flam**, and A. Vishwanath, “Entanglement negativity spectrum of random mixed states: A diagrammatic approach,” arXiv:2011.01277 [cond-mat.str-el]. Phys. Rev. X Quantum 2, 030347.
34. **J. Kudler-Flam**, Y. Kusuki, and S. Ryu, “The quasi-particle picture and its breakdown after local quenches: mutual information, negativity, and reflected entropy,” arXiv:2008.11266 [hep-th]. Journal of High Energy Physics 03 146 (2021).
35. **J. Kudler-Flam**, M. Nozaki, S. Ryu, and M. Tian Tan, “Entanglement of Local Operators and the Butterfly Effect,” arXiv:2005.14243 [hep-th]. Phys. Rev. Res. 3, 033182.
36. M. Asrat and **J. Kudler-Flam**, “$T\bar{T}$, the entanglement wedge cross section, and the breakdown of the split property,” arXiv:2005.08972 [hep-th]. Phys. Rev. D 102, 045009.
37. I. MacCormack, M. Tian Tan, **J. Kudler-Flam**, and S. Ryu, “Operator and entanglement growth in non-thermalizing systems: many-body localization and the random singlet phase,” arXiv:2001.08222 [cond-mat.str-el]. Phys. Rev. D 104, 214202.
38. **J. Kudler-Flam**, Y. Kusuki, and S. Ryu, “Correlation measures and the entanglement wedge cross-section after quantum quenches in two-dimensional conformal field theory,” arXiv:2001.05501 [hep-th]. Journal of High Energy Physics 04, 74 (2020).
39. **J. Kudler-Flam**, L. Nie, and S. Ryu, “Conformal field theory and the web of quantum chaos diagnostics,” arXiv:1910.14575 [hep-th]. Journal of High Energy Physics 01, 175 (2020).
40. **J. Kudler-Flam**, H. Shapourian, and S. Ryu, “The negativity contour: a quasi-local measure of entanglement for mixed states,” arXiv:1908.07540 [hep-th]. SciPost Phys. 8, 063 (2020).
41. Y. Kusuki, **J. Kudler-Flam**, and S. Ryu, “Derivation of holographic negativity in AdS_3/ CFT_2,” arXiv:1907.07824 [hep-th]. Phys. Rev. Lett. 123, 131603.
42. **J. Kudler-Flam**, M. Nozaki, S. Ryu, and M. Tian Tan, “Quantum vs. classical information: operator negativity as a probe of scrambling,” arXiv:1906.07639 [hep-th]. Journal of High Energy Physics 01, 031 (2020).
43. **J. Kudler-Flam**, I. MacCormack, and S. Ryu, “Holographic entanglement contour, bit threads, and the entanglement tsunami,” arXiv:1902.04654 [hep-th]. J. Phys. A: Math. Theor. 52 (2019) 325401.
44. **J. Kudler-Flam** and S. Ryu, “Entanglement negativity and minimal entanglement wedge cross-sections in holographic theories,” arXiv:1808.00446 [hep-th]. Phys. Rev. D 99, 106014.

45. J. Chen and **J. Kudler-Flam**, "Laplacian growth & sandpiles on the Sierpinski gasket: limit shape universality and exact solutions," arXiv:1807.08748 [math-ph]. Annales de l'Institut Henri Poincaré D 7 (4).

Invited Talks

1. Simons Center for Geometry and Physics, Stony Brook, NY — May 2026
   Workshop on DSSYK Model: From Gravity to Many-Body Quantum Chaos
   Wormholes and Averaging over N
2. Stanford University, Stanford, CA — Mar. 2026
   SITP Monday Colloquium
   Large fluctuations in the large-N limit
3. OIST, Okinawa, Japan — Mar. 2026
   Qubits and Spacetime Unit Group Seminar
   Generalized Entropy is von Neumann Entropy
4. University of Michigan, Ann Arbor, MI — Mar. 2026
   LITP Workshop on Quantum Black Holes
   Large fluctuations in the large-N limit
5. Institute for Advanced Study, Princeton, NJ — Dec. 2025
   Workshop on Quantum Aspects of Black Holes and Spacetime
   Emergent Mixed States for Baby Universes and Black Holes
6. Steklov Mathematical Institute, Moscow, Russia — Nov. 2025
   Frontiers in Holography Workshop
   Emergent Mixed States for Baby Universes and Black Holes
7. Massachusetts Institute of Technology, Cambridge, MA — Nov. 2025
   String/Gravity Theory Seminar
   Stringy algebras, stretched horizons, and quantum-connected wormholes
8. Brandeis University, Waltham, MA — Oct. 2025
   Quantum/Gravity Seminar
   Emergent Mixed States for Baby Universes and Black Holes
9. University of Massachusetts, Amherst, MA — Oct. 2025
   Amherst Center for Fundamental Interactions Seminar
   Emergent Mixed States for Baby Universes and Black Holes
10. Institute for Advanced Study, Princeton, NJ — Oct. 2025
    Quantum Aspects of Black Holes Meeting
    Emergent Mixed States for Baby Universes and Black Holes
11. Institute for Advanced Study, Princeton, NJ — Oct. 2025
    Quantum Aspects of Black Holes Meeting
    Discussion of the Lorentzian Gravitational Path Integral
12. Castello dei Principi Capano, Pollica, Italy — Sep. 2025
    Workshop on Universality in non-equilibrium quantum matter
    An Optimal Bound on Long-Range Distillable Entanglement
13. Perimeter Institute, Waterloo, Canada — Jun. 2025
    QIQG 2025
    Absolute Entropy and the Observer's No-Boundary State
14. Institute for Advanced Study, Princeton, NJ — Jun. 2025
    Quantum Aspects of Black Holes Meeting
    Absolute Entropy and the Observer's No-Boundary State
15. Princeton University, Princeton, NJ — Mar. 2025
    PCTS Year End Celebration
    Comments on the identity matrix

16. Texas Tech, Lubbock, TX — Apr. 2025
    Quantum Information and Computing Seminar
    An optimal bound on long-range distillable entanglement
17. Brown University, Providence, RI — Apr. 2025
    Brown Theoretical Physics Center IDEA Seminar Series
    Entropy and the Hartle-Hawking State
18. Cornell University, Ithaca, NY — Feb. 2025
    LEPP Theory Seminar
    Aspects of the Hartle-Hawking State
19. Institute for Advanced Study, Princeton, NJ — Feb. 2025
    Physics Group Meeting
    An Energy Efficient way to send Qubits into Black Holes
20. University of Massachusetts, Amherst, MA — Jan. 2025
    NE Gravity Workshop
    Black Hole Entropy is von Neumann Entropy
21. Institute for Advanced Study, Princeton, NJ — Nov. 2024
    Quantum Aspects of Black Holes Meeting
    Vacua and Infrared Tails in de Sitter
22. Trinity College, Dublin, Ireland — June. 2024
    HMI Workshop: CFT and Holography for Heavy and Thermal States and Black Holes
    Black Hole Entropy is von Neumann Entropy
23. University of Toronto, Toronto, Canada — May. 2024
    Theoretical High-Energy Physics Seminar
    Algebraic Observational Cosmology
24. University of Chicago, Chicago, IL — Mar. 2024
    Quantum Aspects of Cosmology Workshop
    Algebraic Observational Cosmology
25. Institute for Advanced Study, Princeton, NJ — Mar. 2024
    Quantum Aspects of Black Holes Meeting
    Algebraic Observational Cosmology
26. Caltech University, Pasadena, CA — Mar. 2024
    High Energy Theory Seminar
    Algebras and entropies for quantum field theory, black holes, and inflationary cosmology
27. Institute for Advanced Study, Princeton, NJ — Oct. 2023
    Physics Group Meeting
    Entropy differences in quantum field theory and a rigorous proof of the Bekenstein bound
28. KTH Royal Institute of Technology, Stockholm, Sweden — May 2023
    Complex Systems and Biological Physics Seminars
    Horizons as eavesdroppers: decoherence from soft radiation
29. Institute for Basic Science, Daejeon, Republic of Korea — May 2023
    Center for Theoretical Physics of Complex Systems Seminar
    Non-Hermitian Hamiltonians Violate the Eigenstate Thermalization Hypothesis
30. Purdue University, West Lafayette, IN — Apr. 2023
    Department of Physics & Astronomy Seminar
    Horizons as eavesdroppers: decoherence from soft radiation
31. Oxford University, Oxford, UK — Apr. 2023
    Sondhi Group Meeting
    Horizons as eavesdroppers: decoherence from soft radiation

32. University of California Berkeley, Berkeley, CA — Apr. 2023
Berkeley Center for Theoretical Physics String Seminar
Horizons as eavesdroppers: decoherence from soft radiation

33. Stanford University, Stanford, CA — Apr. 2023
Stanford Institute for Theoretical Physics Seminar
Horizons as eavesdroppers: decoherence from soft radiation

34. City College of the City University of New York, New York, NY — Mar. 2023
Theoretical Physics Seminar
Horizons as eavesdroppers: decoherence from soft radiation

35. Institute for Advanced Study, Princeton, NJ — Mar. 2023
Quantum Aspects of Black Holes Meeting
The Information-Disturbance Tradeoff and Horizons as Eavesdroppers

36. University of Pennsylvania, Philadelphia, PA — Feb. 2023
High Energy Theory Seminar
Rényi mutual information in quantum field theory and gravity

37. Southeast University, Nanjing, China — Dec. 2022
2022 Forum for Young Scholars in Mathematics and Mathematical Physics
Rényi mutual information in quantum field theory

38. Institute for Advanced Study, Princeton, NJ — Sep. 2022
High Energy Theory Seminar
Quantum Information before the Page Time

39. Princeton University, Princeton, NJ — Sep. 2022
Princeton Center for Theoretical Science Annual Retreat
Distinguishability and its use: from spin chains to black holes

40. University of Amsterdam, Amsterdam, Netherlands — Mar. 2022
Institute of Physics String Theory Seminar
On the Information Content of Black Hole Radiation

41. Steklov Mathematical Institute, Moscow, Russia — Dec. 2021
Frontiers in Holography Workshop
Distinguishability in Random States, Eigenstates, and Gravity

42. Institute for Research in Fundamental Sciences, Tehran, Iran — Nov. 2021
IPM Holography Virtual Seminar
Distinguishability in Random States, Eigenstates, and Gravity

43. University of Illinois, Urbana Champaign, IL — Oct. 2021
Mathematical and Theoretical Physics Seminar
Distinguishability in Random States, Eigenstates, and Gravity

44. Harvard University, Cambridge, MA — Jul. 2021
Vishwanath Group Meeting
Distinguishability of random states and the subsystem ETH

45. Strings 2021 Gong Show, São Paulo, Brazil — Jun. 2021
Distinguishing Random States and Black Holes

46. Yukawa Institute for Theoretical Physics, Kyoto, Japan — Feb. 2021
Entanglement Meeting
Entanglement Negativity Spectrum of Random Mixed States

47. University of Illinois Urbana Champaign, Champaign IL — Oct. 2020
Condensed Matter-AMO Journal Club
Universality of Entanglement Dynamics in Integrable and Chaotic Quantum Systems

48. Massachusetts Institute of Technology, Cambridge, MA — Aug. 2020
CTP Group Meeting
$T\bar{T}$, the entanglement wedge cross section, and the breakdown of the split property

49. Stanford University, Stanford, CA — Jun. 2020
SITP $T\bar{T}$ Group Meeting Seminar
$T\bar{T}$, the entanglement wedge cross section, and the breakdown of the split property

50. Colgate University, Hamilton, NY — Jun. 2020
Physics & Astronomy Seminar
Entanglement, Gravity, and the Holographic Principle

51. Massachusetts Institute of Technology, Cambridge, MA — Apr. 2020
Condensed Matter Theory Seminar
Universality of Entanglement Dynamics

52. Albert Einstein Institute, Potsdam, Germany — Dec. 2019
Gravity, Quantum Fields, & Information Seminar
Derivation of holographic negativity in AdS_3/CFT_2

53. AMS Special Sessions on Analysis and Geometry of Fractals, Riverside, CA — Nov. 2017
Strong shape theorems in cellular automata models on fractal graphs

Teaching Experience

Prison Teaching Initiative Instructor — 2023-Present
Princeton University, Princeton, NJ

- **PHYS-130**: *Astronomy* (Fall 2025) East Jersey State Prison
- **PHYS-130**: *Astronomy* (Spring 2025) Northern State Prison
- **PHYS-130**: *Astronomy* (Fall 2024) East Jersey State Prison
- **MATH-110**: *Statistics* (Spring 2024) South Woods State Prison
- **MATH-015**: *Basic Mathematics* (Fall 2023) South Woods State Prison

Prison Teaching Initiative Course Coordinator — 2024-2025
Princeton University, Princeton, NJ

- **PHYS-130**: *Astronomy* (Spring 2025)
- **PHYS-130**: *Astronomy* (Fall 2024)

Guest Lecturer — 2024
Princeton University Department of Physics, Princeton, NJ

- **PHY 101**: *Introductory Physics I.* Lecture on "Vibrational Motion and Mechanical Waves."

Tyler School of Art and Architecture, Philadelphia, PA

- **ART 3505**: *Color.* Lecture on "The Physics of Light and Color."

Princeton University Teaching Transcript Program — 2022-2024
The McGraw Center for Teaching and Learning, Princeton University, Princeton, NJ

- Includes five pedagogy workshops, guest lecturing with classroom observation and analysis, and the development of an original syllabus. Completed Fall 2024.

Reading Course Instructor — 2019
University of Chicago, Chicago, IL

- Designed and taught reading course on *"Quantum Information and Quantum Chaos"* to undergraduate physics majors at University of Chicago.

Splash! Chicago Teacher — 2018
University of Chicago, Chicago, IL

- Designed and taught a (5 week) course *"Introduction to Quantum Mechanics"* to local high school students.
- Designed and taught a (1 day) course *"Introduction to Quantum Mechanics and Quantum Computing"* to local high school students.

Graduate Teaching Assistant 2017 - 2021
Department of Physics, University of Chicago, Chicago, IL

- **PHSC 116**: *Physics for Future Presidents: Fundamental Concepts*,
- **PHSC 117**: *Physics for Future Presidents: Energy & Sustainability*,
- **PHYS 121-123**: *General Physics I-III*
- **PHYS 133**: *Waves, Optics, & Heat*
- **PHYS 142**: *Honors Electricity & Magentism*
- **PHYS 341**: *Graduate Quantum Mechanics I* (Grader)
- **PHYS 484**: *String Theory II* (Grader)

Undergraduate Teaching Assistant and Tutor 2015-2017
Department of Physics & Astronomy, Colgate University, Hamilton, NY

- **PHYS 232**: *Introduction to Mechanics*
- **CORE 101**: *Energy and Sustainability*
- **PHYS 111**: *Fundamental Physics I*
- **ASTR 102**: *Stars, Galaxies, and the Universe*

Mentorship

Research Advisor
Bowen Ouyang - 4^{th} year undergraduate student at Shanghai Jiao Tong University (2025)
Hyaline Chen - 3^{rd}-4^{th} year undergraduate Physics student at Princeton University (2024). Resulting publication. Winner of PACM Undergraduate Certificate Prize for Best Research Project
Chang-Han Chen - 4^{th} year undergraduate Physics student at Massachusetts Institute of Technology (2020-2021). Senior Thesis

Professional Activities

Journal Referee
Physical Review Letters, Physical Review E, Nature Physics, Journal of High Energy Physics, SciPost Physics, European Physical Journal Plus, European Physical Journal C, Journal of Mathematical Physics, Quantum

Organizer
Institute for Advanced Study/Princeton Center for Theoretical Science Workshop: "Workshop on Quantum Aspects of Black Holes and Spacetime" (Dec 2025)
Princeton Center for Theoretical Science Workshop: "Random Physics" (May 2024)
Princeton Center for Theoretical Science Lunch Seminars (Fall 2023-Spring 2024)
Princeton Center for Theoretical Science Workshop: "Quantum Information, Dynamics and Ergodicity: From Many-Body Systems to Gravity" (Feb.-Mar. 2023)
Kadanoff Center for Theoretical Physics Journal Club (University of Chicago)
Kadanoff Center Group Meeting → Theory Group Meeting (University of Chicago)